

8. There are many different types of cancer. One of the most common is breast cancer. In approximately 30% of breast cancers a high level of HER2 protein is present, this is due to a mutation in the HER2 gene. The high level of HER2 protein found on the surface of the cells causes them to grow and divide excessively.

(a) State what is meant by the term mutation. [1]

.....

.....

(b) (i) In Wales, approximately 2900 people are diagnosed with breast cancer each year. Calculate how many of those diagnosed would be expected to have a high level of HER2 protein present. [2]

Number of people =

(ii) Explain how the HER2 gene codes for the production of HER2 protein. [3]

.....

.....

.....

.....

.....

.....



Examiner
only

- (c) Since 1985, approximately 100 different monoclonal antibodies have been developed and some are used in the treatment of breast cancer linked to the mutated HER2 gene. During chemotherapy a drug is attached to the monoclonal antibody to target these cancer cells directly.

Explain how the monoclonal antibody targets these cancer cells directly. [3]

.....

.....

.....

.....

.....

.....

- (d) State **two** other uses of monoclonal antibodies. [2]

.....

.....

11

